

AMIT SINGH

Bhubaneswar, Odisha - 751024, India

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EDUCATION

- **KIIT (Kalinga Institute of Industrial Technology)** *Bhubaneswar, India*
B.Tech. in Computer Science and Engineering, CGPA: 8.52/10
2023 – 2027
- **Secondary Education (+2)** *Birgunj, Nepal*
Science Stream, CGPA: 3.20/4
Himalayan Pyramid +2 School
2021 – 2023

TECHNICAL SKILLS

- **Programming Languages:** C, Python, Java
- **Database Systems:** MySQL, MongoDB
- **AI & Machine Learning:** NumPy, Pandas, Matplotlib, Seaborn, PyTorch, Scikit-learn, TensorFlow, Keras
- **DevOps & Version Control:** Git, GitHub, Docker, CI/CD

PROJECTS

- **Real Time Emotions Detector**
Python, TensorFlow, Keras, OpenCV, JavaScript
 - Developed an AI-based emotion recognition system using DeepFace to detect 7 human emotions (happy, sad, angry, surprised, fearful, disgusted, neutral) from facial images with (90-95%) accuracy.
 - Built a full-stack web application using Python (Flask) and HTML/CSS, exposing REST APIs for real-time image-based emotion detection.
 - Integrated pre-trained deep learning models enabling zero-training deployment and optimized lightweight frontend for fast and user-friendly interaction.
- **Hostel Management System**
MongoDB, Express.js, Node.js, JavaScript
 - Developed and deployed a full-stack Hostel Management System using MongoDB, Express.js, Node.js, and JavaScript with secure JWT-based authentication and role-based access (Owner, Admin, Student).
 - Built scalable RESTful APIs and an interactive admin dashboard for student management, room allocation, attendance, complaints, and hostel operations; deployed on Render with MongoDB Atlas and bcrypt security.
- **DermAI: Deep Learning-Based Acne Detection with Explainable AI (XAI)**
 - Developed DermAI, a clinical AI web app using EfficientNet-B0 (PyTorch) achieving **91.6% accuracy** for real-time acne detection with a Tri-Layer XAI framework (Grad-CAM, LIME, SHAP).
 - Built a full-stack system using Python (Flask) and Vanilla JS, integrating Groq/OpenAI APIs for AI-driven skincare insights with a responsive UI.
 - Deployed on Netlify + Render, enabling real-time predictions, scan history tracking, and personalized skincare recommendations.

PUBLICATIONS AND CERTIFICATIONS

- [1] Amit Singh (2026). **Explainable AI for Preclinical Risk Detection of Bile Duct Cancer Using Multi-Omics, Microbiome Data and Clinical Biomarkers**. In *Proceedings of the International Symposium on Embedded Systems and Devices (ISED)*, pp. 6.0. Date: 27 February 2026, Location: Raipur, India. DOI: 10.1109/ISED67359.2025.11405038
 - Certificate of presentation: [Link](#)